Low-Cost Motion Tracking in the Browser

Synchronizing IMU signals with video pose estimation for accessible motion inspection

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Motivation

- Commercial motion capture is often expensive and requires dedicated hardware.
- Clinics, rehab, sports, and ergonomics need quick, repeatable motion inspection.
- Combining video context with IMU peaks improves interpretability.

Aim & Research Question

Aim: Build a low-cost, low-complexity prototype that runs in a standard web browser.

Research question: How can IMU data and video-based pose estimation be integrated and synchronized for motion analysis?

Key Requirements

- Runs locally in a standard browser (no server).
- Import IMU recordings (CSV) and uploaded video.
- Show pose skeleton overlay + IMU plots side-by-side.
- Provide practical synchronization and basic annotations.

System Overview

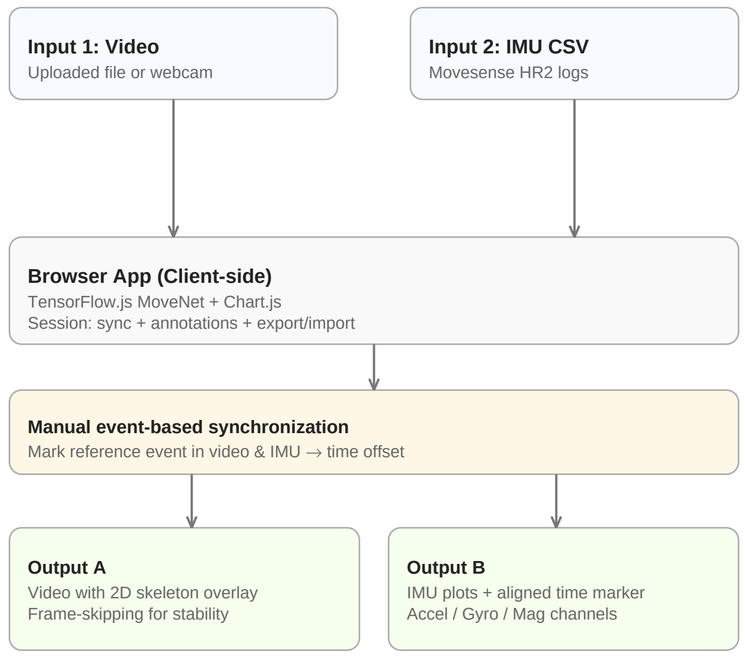


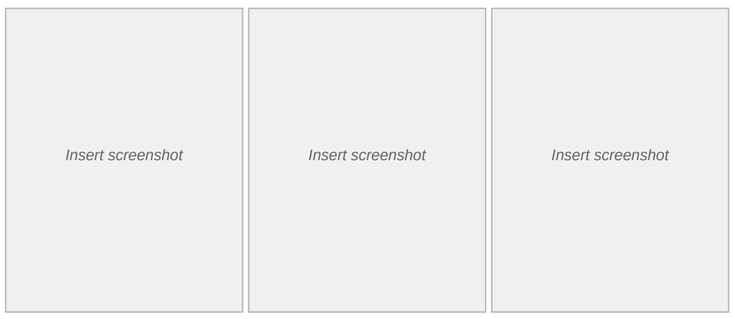
Figure: Data flow and core modules of the IMU-video motion inspection prototype.

Methods (What we built)

- Pose estimation: MoveNet Thunder for uploaded videos; Lightning for webcam mode.
- IMU handling: CSV import, timestamps reconstructed from fixed 104 Hz sampling.
- Visualization: skeleton overlay on canvas; three stacked IMU time-series with sliding window.
- Session tools: local storage, export/import (ZIP), timestamp annotations, TXT report.

Results (Prototype functions)

- Uploaded video playback with 2D skeleton overlay.
- IMU CSV plotted as accelerometer, gyroscope, magnetometer charts.
- Manual markers compute a sync offset; IMU window follows video time.
- Annotations (label, type, notes) and a TXT summary report.



Skeleton overlay | Synchronized IMU plots | Annotations & export

Limitations

- 2D pose keypoints: sensitive to occlusion and camera viewpoint.
- No drift compensation; sync assumes constant time offset.
- Pose inference and playback stability vary across devices.
- IMU timing depends on assumed sampling rate and logging stability.

Ethics & Data Handling

- Client-side processing: video and sensor data stay on the user device.
- Testing used only group members; future deployments require informed consent.
- Prototype is not clinically validated; avoid diagnostic use without validation.

Future Improvements

- Auto-sync via detected calibration event (e.g., clap) in video + IMU peak.
- Multiple events to estimate and correct clock drift.
- Add motion metrics and explore sensor fusion (e.g., complementary/Kalman filters).
- Support multiple sensors and multi-person tracking.

Take-home message: A browser-based workflow can enable low-cost exploratory motion inspection using IMU + video.

Reference: TensorFlow MoveNet (blog.tensorflow.org, 2021).